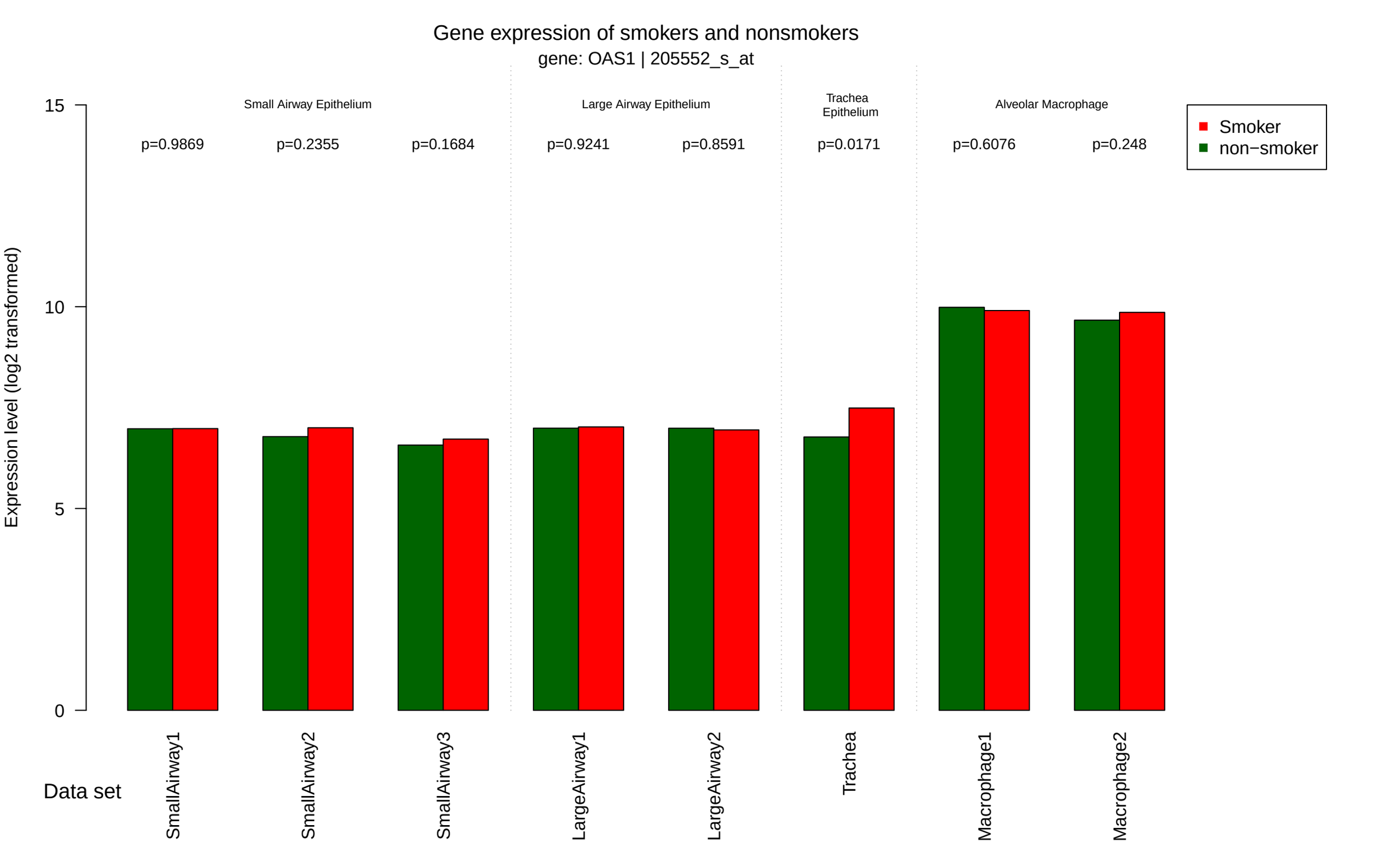
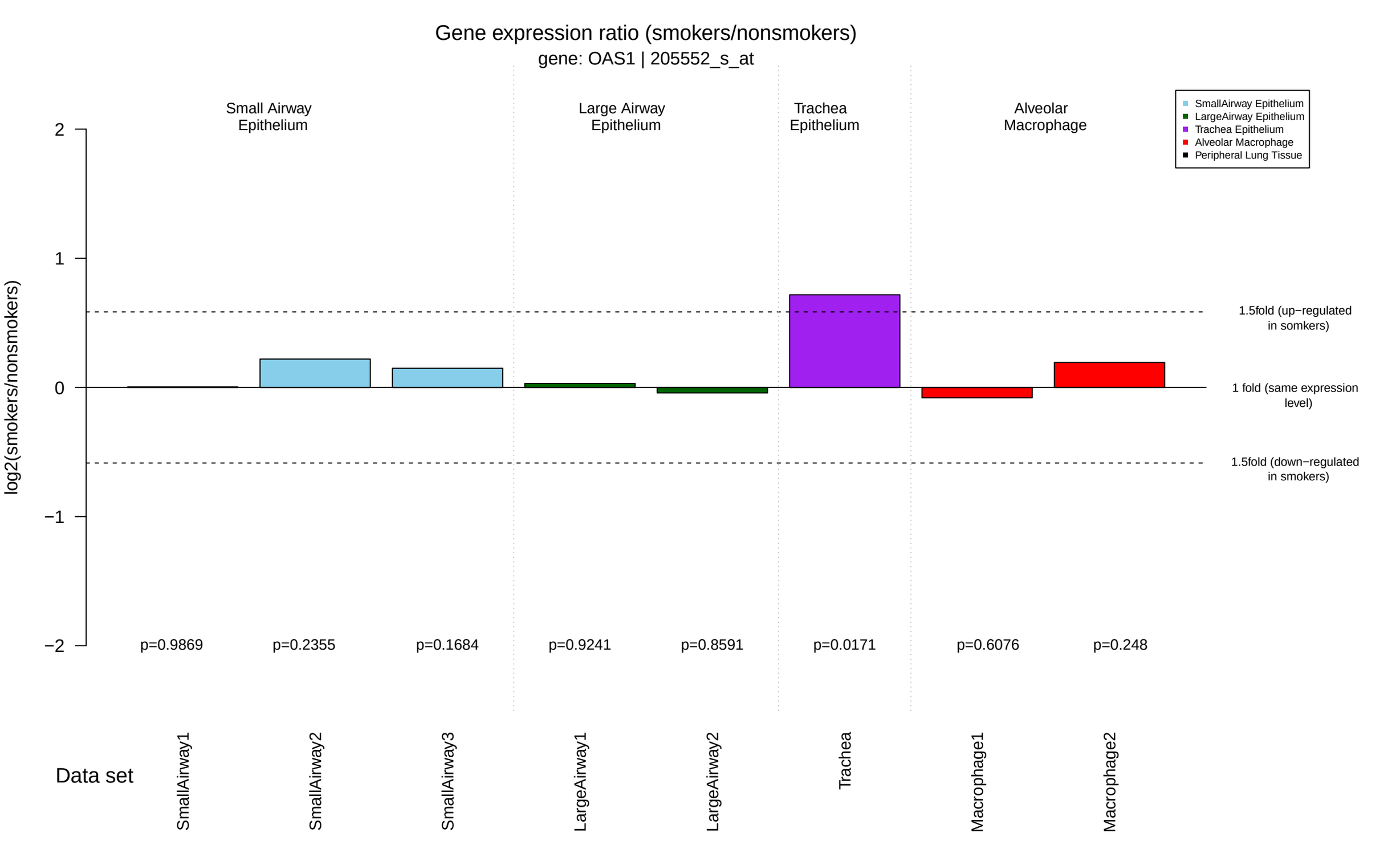
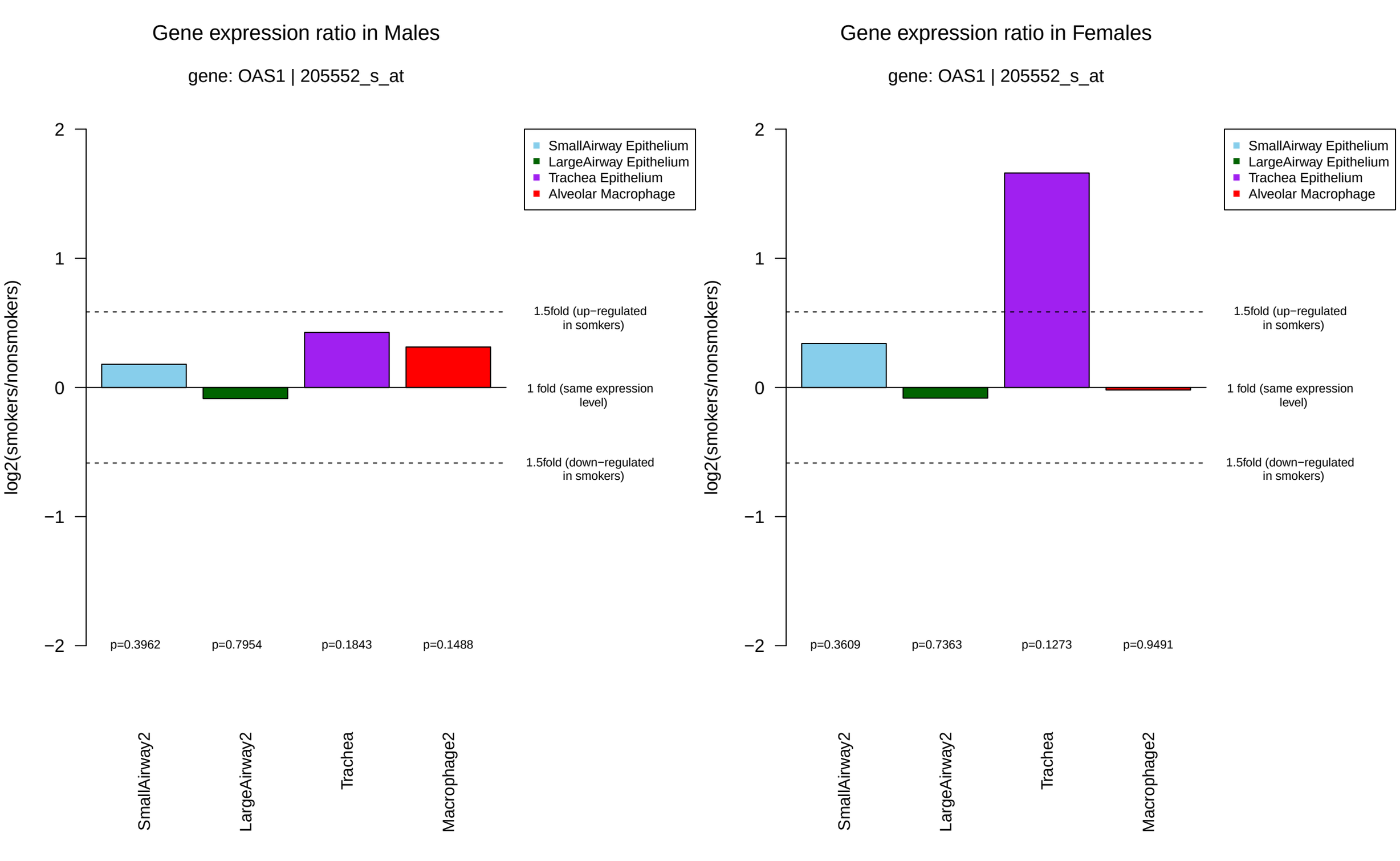
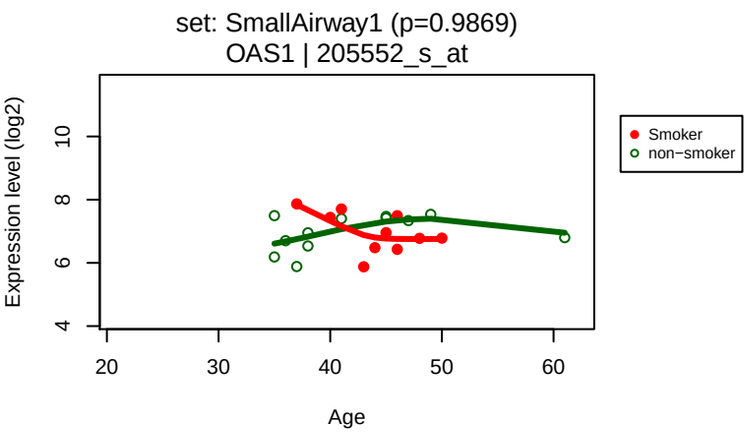
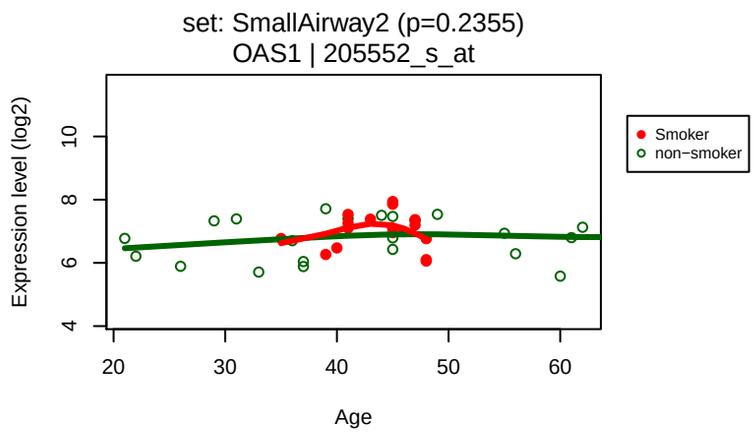
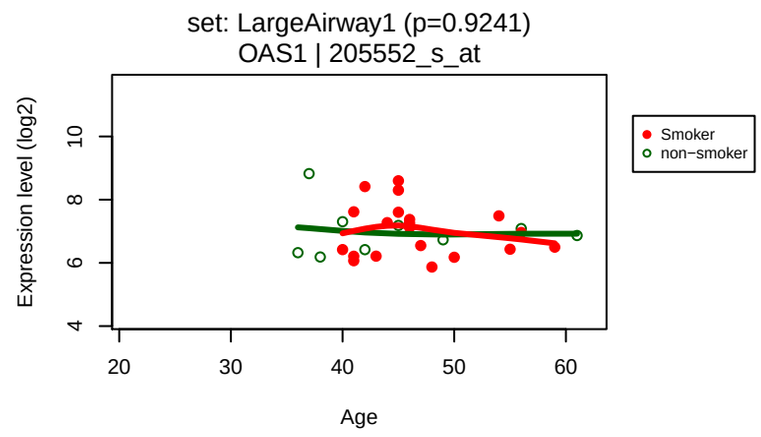

Gene expression of smokers and nonsmokers
gene: OAS1 | 205552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9869
p=0.2355
p=0.1684
p=0.9241
p=0.8591
p=0.0171
p=0.6076
p=0.248
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

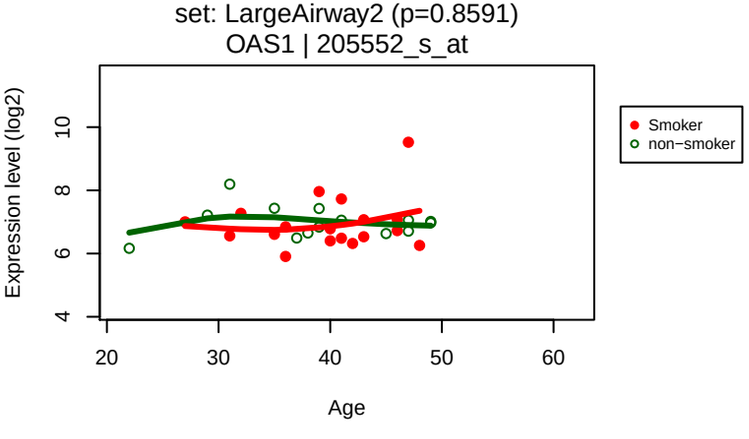

Gene expression ratio (smokers/nonsmokers)
gene: OAS1 | 205552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9869
p=0.2355
p=0.1684
p=0.9241
p=0.8591
p=0.0171
p=0.6076
p=0.248
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

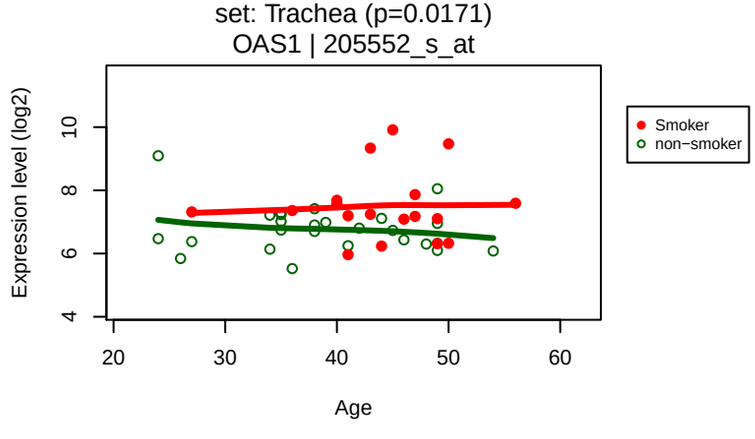

Gene expression ratio in Males
gene: OAS1 | 205552_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3962
p=0.7954
p=0.1843
p=0.1488
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OAS1 | 205552_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3609
p=0.7363
p=0.1273
p=0.9491
SmallAirway2
LargeAirway2
Trachea
Macrophage2

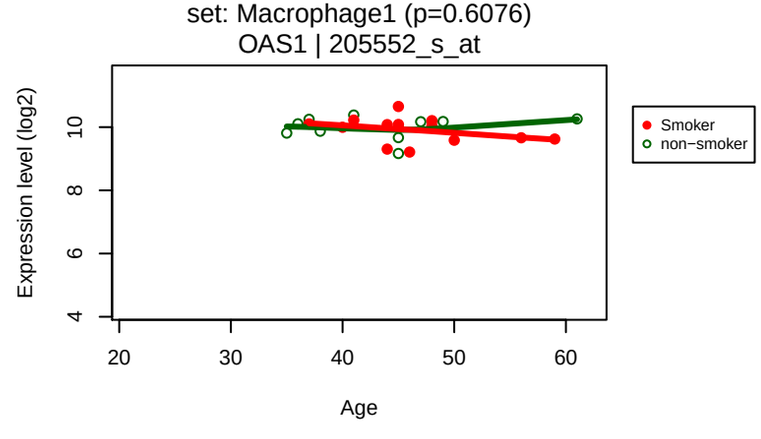

set: SmallAirway1 (p=0.9869)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker

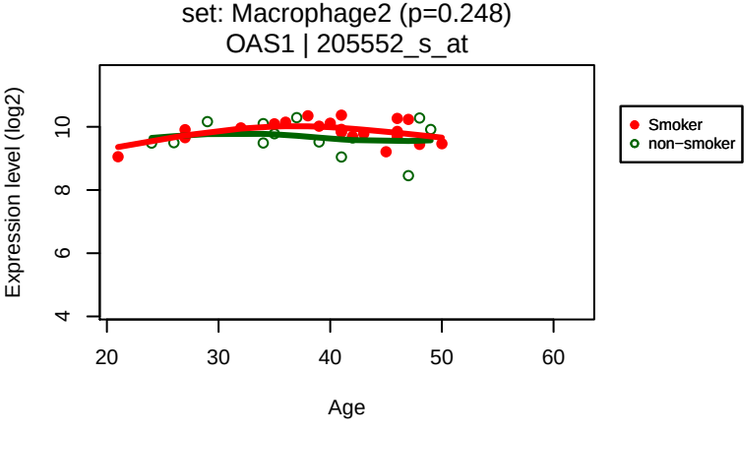

set: SmallAirway2 (p=0.2355)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker

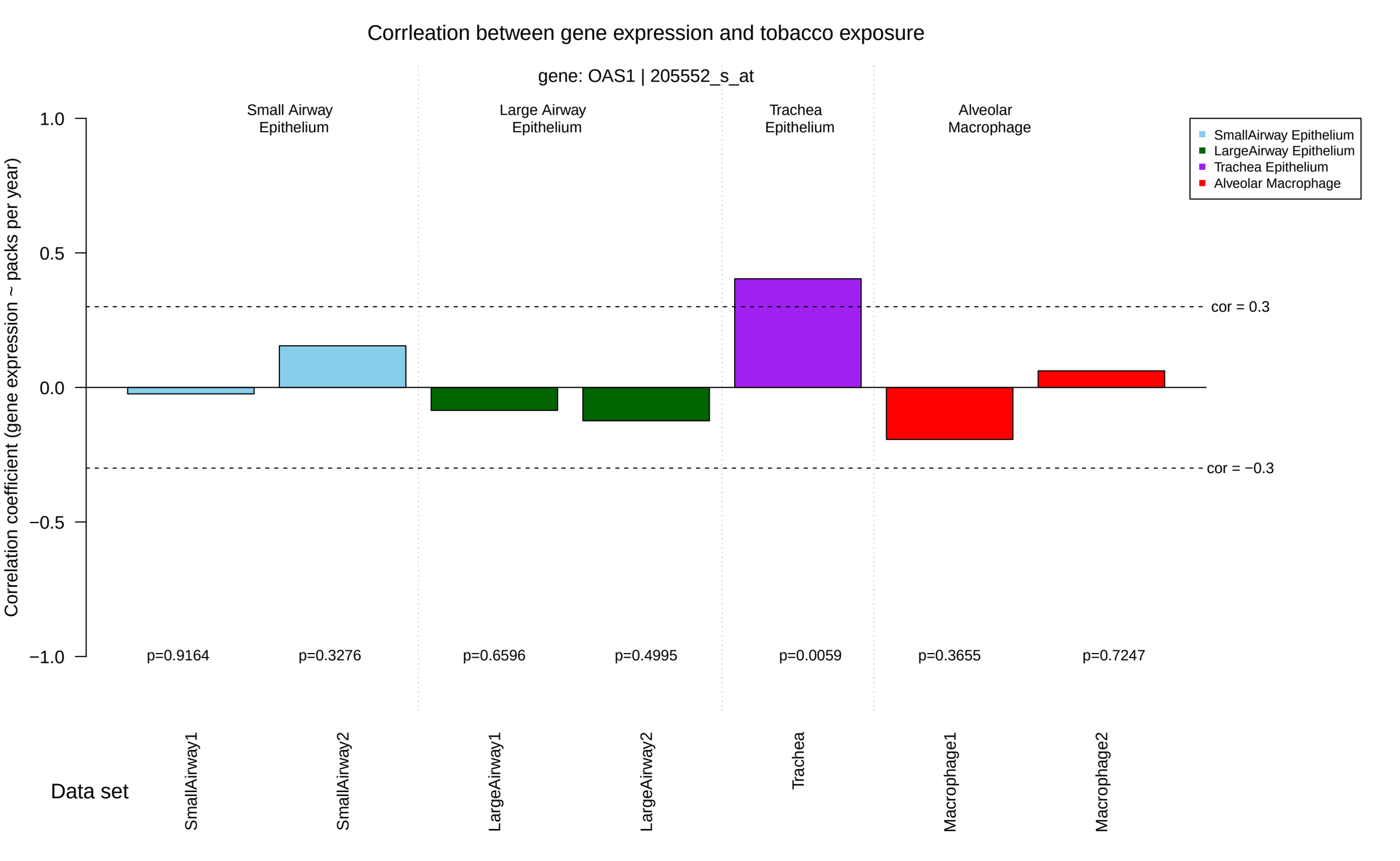

set: LargeAirway1 (p=0.9241)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8591)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0171)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6076)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.248)
OAS1 | 205552_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OAS1 | 205552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9164
p=0.3276
p=0.6596
p=0.4995
p=0.0059
p=0.3655
p=0.7247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2